

Enhancing Cybersecurity in Nigeria: A Proposed Risk Management Framework for Universities

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Abstract— Amid the constant evolution of the digital landscape, cybercrime poses a global threat that looms over Nigeria. Within this context, universities in Nigeria confront vulnerabilities in the realm of cyberspace governance. Consequently, it becomes imperative for these Nigerian educational institutions to embrace a meticulously crafted, all-encompassing framework for managing the risks posed by cybercrime. This step is essential to fortify their defenses, uphold their integrity, preserve confidentiality, and maintain their esteemed reputation. Tragically, the incursion of cyber threats and breaches exacts far-reaching consequences—financial, societal, and concerning intellectual property—upon these esteemed academic entities. This occurs despite their earnest endeavors to invest in measures designed to counteract cybercrime. This sobering reality underscores the necessity for tailored and scientifically informed frameworks, capable of thwarting both internal and external cyber threats. Confronting this pressing quandary, this study proposes actionable frameworks that draw insights from existing literature while integrating state-of-the-art tools such as intrusion detection systems, advanced cryptography, and threat analytics driven by artificial intelligence. Guided by a rigorous process encompassing problem recognition, objective establishment, artifact formulation, meticulous testing, comprehensive assessment, and transparent communication of outcomes, Nigerian universities can establish robust cybersecurity initiatives. These frameworks serve as a promising foundational stride, sheltering sensitive data, intellectual property, and reputational capital, fostering an environment of cyber resilience conducive to undeterred academic pursuits. To ensure sustained cybersecurity resilience within academic institutions, adaptability to the unpredictable landscape of cybercrime remains paramount. The ongoing refinement and flexibility of these frameworks empower them to withstand the ceaseless cat-and-mouse engagement between cyber malefactors and defenders, bolstering the digital boundaries of Nigerian universities against the unrelenting deluge of cyber threats.

Keywords— *Cybersecurity, cybercrime mitigation, risk management framework, sensitive data, Nigerian Universities*

I. INTRODUCTION

In today's world of digital connectivity, it is crucial to prioritize the enhancement of cybersecurity measures in Nigeria's universities. Keeping our institutions safe from cyber threats is of utmost importance, and proactive steps ensure that data and information remain secure [1]. As

institutions of higher learning increasingly rely on digital technologies for education and administrative processes, they become lucrative targets for cyberattacks. To address these threats and safeguard sensitive data, a robust risk management framework tailored to the unique needs of universities is essential. This proposed framework aims to provide a comprehensive approach to cybersecurity, emphasizing prevention, detection, response, and recovery strategies, while also promoting a culture of cybersecurity awareness and resilience within Nigerian universities [2].

According to [2], cyberspace is the interconnected digital environment, encompassing websites, databases, and cloud services. It enables global connectivity and e-learning, fostering research and administrative efficiency. [3] opined that cybersecurity measures, like firewalls and encryption, protect sensitive data. Therefore, communication tools such as websites and email enhance outreach while resource sharing and data analytics optimize collaboration and decision-making [3,1]. Because it offers cost-efficient solutions through digital transformation, cyberspace is pivotal for Nigerian universities, promoting connectivity, e-learning, research, security, and administrative efficiency while fostering digital transformation and resource sharing [2].

The evolution of cyberspaces has been nothing short of transformative, especially in the context of Nigerian universities. These digital realms have grown and diversified, providing a myriad of platforms and tools that universities now rely on for a multitude of critical functions [4]. Cyberspaces have become the cornerstone of modern education enabling universities to extend their reach far beyond physical campuses. Through e-learning platforms and online courses, students can access educational materials, engage in lectures, and collaborate with peers and instructors from virtually anywhere, breaking down geographical barriers and making education more accessible [3-1].

In the realm of research, cyberspaces have revolutionized the way universities conduct studies and share findings [4]. [4] opined that online libraries, databases, and research platforms grant researchers access to a wealth of knowledge and resources, accelerating the pace of discovery. In today's interconnected world, it is increasingly common for institutions to collaborate, even across continents. This allows for the valuable sharing of ideas and expertise on a global scale. Transcending from physical attendance, community development within universities has also been greatly

influenced by cyberspaces—social media, forums, and online communities provide platforms for students, faculty, and alumni to connect, share experiences, and support each other's academic and professional journeys [4]. As cyberspaces continue to evolve, they offer opportunities for Nigerian universities to further enhance their teaching, research, community engagement, and administrative capabilities. Embracing these digital platforms and staying abreast of emerging technologies will be key to ensuring that universities remain competitive and responsive to the ever-changing needs of the academic landscape.

Recent cybercrime incidents have dramatically expanded our understanding of the scope of cyber threats, extending well beyond the realm of financial institutions. Notably, [5,3,4] have shed light on the alarming surge in cyberattacks against universities, positioning the education sector as the second most targeted industry for cybersecurity threats. These threats resonate globally, exemplified by the FBI and CISA's warning of state-sponsored hackers targeting organizations, particularly universities, involved in critical COVID-19 research [6]. Reports from multiple countries echo a concerning trend of cyber espionage, emphasizing the urgency of fortified cybersecurity measures within academic institutions, where valuable intellectual property and sensitive information are increasingly at risk [1].

In Nigerian universities, for instance, the Federal University of Technology Akure fell victim to a cryptic Denial of Service [DoS] attack, with an unidentified individual exploiting the Network Time Protocol [NTP] server [2]. In a similarly nebulous vein, Madonna University experienced a significant data breach, compromising over 25,000 records within its database . Ahmadu Bello University in Zaria unveiled a series of breaches between 2016 and 2018, casting shadows over the security of student and staff data [2]. Additionally, there were enigmatic cases of university staff manipulating admission records and room allocations, underscoring the multifaceted and intricate nature of cybersecurity issues faced by these institutions [4].

[5] described a Risk Management Framework [RMF] for universities as a structured cybersecurity methodology that employs technical terminologies and practices to systematically identify, assess, and mitigate risks in the university's digital landscape. This framework entails the meticulous identification of vulnerabilities, threats, and critical assets within the university's network [6], followed by a rigorous risk assessment that quantifies the likelihood and potential impact of these risks [2]. Subsequently, it involves the implementation of robust cybersecurity measures, including access controls, encryption protocols, intrusion detection systems, and security training programs [6]. Therefore, RMF integrates prevention, detection, response, and recovery strategies to fortify the security posture, ensuring the protection of sensitive data, systems, and academic operations while addressing the dynamic landscape of evolving cyber threats [6,1,2].

By implementing an RMF, [5] proposed this framework involves identifying potential vulnerabilities, assessing risks, and prioritizing strategies for prevention, detection, response, and recovery. [6] add that it includes measures such as regular risk assessments, robust access controls, user awareness

programs, real-time monitoring, incident response planning, and collaboration with cybersecurity organizations. This approach ensures proactive mitigation of threats, fosters a culture of cybersecurity, and strengthens the university's ability to safeguard sensitive data, systems, and academic processes from cyber threats [7].

II. LITERATURE REVIEW

1.1 *Cybercrime and Cybercrime Threats*

According to [1], cybercrime, in essence, refers to criminal activities conducted in the digital realm encompassing a wide range of offenses, including hacking, identity theft, fraud, spreading malware, phishing, and more. Cyber threats are potential risks posed by malicious actors exploiting vulnerabilities for financial gain or unauthorized access. These activities exploit vulnerabilities in computer systems, networks, and electronic devices, posing significant threats to individuals, organizations, and nations alike [2].

Cybercrime's impact on universities necessitates comprehensive responses. These include regulatory frameworks [3], collaborative networks [2], training, incident response plans [4], technological investments, international cooperation, research, and public-private partnerships [5], all aimed at mitigating threats and securing the digital landscape of higher education [1].

The importance of addressing cybercrime within universities is profound [6]. As [5] described, universities' increasing reliance on digital technology has resulted in a significant rise in cybercrime threats. These threats jeopardize the integrity, confidentiality, and availability of critical information, including personal data, research findings, financial records, and intellectual property [2]. Therefore, cyberattacks can disrupt academic activities, compromise student and staff records, and damage an institution's reputation [6, 2].

Globally, responses to cybercrime within HEIs have evolved to counter these multifaceted threats. Governments worldwide have enacted cybersecurity regulations and data protection laws, imposing stringent requirements on universities to safeguard sensitive information [7]. These regulations are critical in holding institutions accountable for their cybersecurity practices.

The escalating threat of cybercrime has indeed positioned it as one of the world's most significant man-made risks, and its implications within universities are far-reaching and demand attention [8, 9]. Data breaches are among the most common cybercrimes affecting universities [9]. For instance, in 2020, the United Kingdom's University of Hertfordshire suffered a significant data breach [10], exposing sensitive information of over 295,000 individuals [11]. Such breaches not only compromise personal data but also incur significant financial and reputational costs for universities.

Ransomware attacks, like the one targeting the University of California, San Francisco [UCSF] in 2020, can paralyze university operations [11]. UCSF paid a ransom of \$1.14 million to regain access to critical research data, showcasing the financial impact of cybercrime [10]. Universities, as hubs of innovation and research, are prime targets for cybercriminals seeking to steal valuable intellectual property [8]. The Australian National University [ANU] experienced a data breach in 2018, resulting in the theft of research data

worth billions of dollars [11]. Additionally, [12] described that cybercrime poses a direct threat to the continuity of academic activities. Distributed Denial of Service [DDoS] attacks can disrupt online learning platforms, preventing students and faculty from accessing essential resources.

Universities have also embraced collaborative networks, facilitating information sharing among academic institutions to enhance cybersecurity [13]. [7] opined those initiatives like the Research and Education Networking Information Sharing and Analysis Center [REN-ISAC] in the United States enable universities to collectively address cyber threats. Their published research and findings can also be accessed and used by other universities around the world for knowledge sharing. Although [14, 8] emphasized the importance of training and awareness programs that can gain prominence within universities, [14] opined educating staff and students about cyber threats and the importance of safe online practices may truncate possible incidents. Incident response plans may need to be established to enable timely responses to cyber incidents, minimizing damage and facilitating recovery [7].

1.2 Cybersecurity Issues in Universities

Cybersecurity is the practice of safeguarding digital assets and information [15]. [1] argues it majorly encompasses ensuring confidentiality, integrity, and availability of data, authentication, vulnerability management, and incident response. However, [4] argues that this concept is vital in protecting against a wide range of threats and vulnerabilities in the digital realm, from data breaches to malware attacks.

According to [2, 9, 12], cybersecurity issues have become a prominent concern in today's interconnected digital landscape. Encompassing a range of threats and vulnerabilities that target information systems, networks, and digital assets [9], universities, as hubs of research, innovation, and sensitive information, are not immune to these challenges [10]. They are particularly susceptible due to the vast amount of personal, financial, and research data they handle [7].

One major cybersecurity issue universities face is data breaches. According to [16], these breaches involve unauthorized access to sensitive information, such as student and faculty records, research data, and financial details. For instance, in 2015, the University of California, Los Angeles [UCLA] experienced a breach where cybercriminals gained access to records of about 4.5 million individuals². This type of breach can lead to identity theft, financial fraud, and reputational damage [2].

Another concern is phishing attacks [14]. According to [5], cybercriminals often send deceptive emails to university members, tricking them into revealing login credentials or clicking on malicious links. In 2016, a major phishing attack targeted staff and students at the University of Central Florida³, compromising login credentials and potentially sensitive data [8]. Phishing attacks result in disrupting university operations, compromising personal data, and even providing hackers with a foothold for more extensive attacks [8, 17].

Ransomware attacks are currently one of the most frequent and harmful cyber threats. These malicious attacks consist of encrypting an organization's data and demanding a ransom for

its recovery. It is a true nightmare for any company, and the best defense against it is prevention [3]. In 2020, the University of California, San Francisco [UCSF] paid a ransom of \$1.14 million after a ransomware attack encrypted critical medical research data⁴. Such attacks can disrupt teaching, research, and administrative functions, as well as result in significant financial losses [11].

In addition, universities tend to harbor numerous IoT [Internet of Things] devices and systems, which can potentially expose security weaknesses [18]. These devices, ranging from connected campus infrastructure to research equipment [14], may lack proper security measures, making them potential targets for hackers seeking to compromise the university's network [2].

The global landscape showcases these cybersecurity challenges in various ways. In 2019, the Australian National University suffered a data breach affecting the personal and financial data of staff, students, and visitors. In the United Kingdom, the University of York experienced a cyberattack in 2020, disrupting its IT services and communication systems⁵. Similarly, Canadian universities have been targeted, with the University of Calgary paying a ransom in 2016 to regain access to its systems [10].

Although the field of cybersecurity is rapidly evolving, scholarly discoveries about cybersecurity in Nigeria has unveiled critical insights into the state of cybercrime within the country. Notably, Nigeria has been identified as a hub for various cybercriminal activities, including phishing, online scams, identity theft, and advance-fee fraud, commonly known as "419" scams [1]. These cybercrimes have resulted in substantial financial losses and widespread reputational damage, both domestically and internationally.

The studies of [19] and [12] illuminated Nigeria's participation in global cybercrime networks as a significant concern. Certain Nigerian cybercriminal groups have gained infamy for their involvement in advanced cyberattacks, such as Business Email Compromise [BEC] scams, ransomware attacks, and various online fraud schemes that target individuals and organizations worldwide⁶.

The challenges facing Nigerian law enforcement agencies and the judiciary in addressing cybercrime have been highlighted in [20] and [21]. These challenges include difficulties in identifying and prosecuting cybercriminals [20], as well as shortcomings in the existing legal framework for combating cybercrime [21]. These limitations necessitate significant improvements in law enforcement capabilities and the development of robust cybercrime laws and enforcement mechanisms [2, 20, 21].

Research has also revealed that a considerable number of cybercriminals in Nigeria are relatively young individuals, attracted to cybercrime due to economic factors, lack of employment opportunities, and the potential for substantial financial gains [1]. Furthermore, international cooperation has been emphasized as an essential component in addressing cybercrime in Nigeria [22, 14, 13]. Given the transnational nature of cyber threats [23], collaborative efforts between Nigerian law enforcement agencies and their international

counterparts are deemed critical for effective cybercrime investigation and prosecution [21].

In response to increasing cyber threats, the Nigerian government has improved its cybersecurity regulations. Research shows that individuals and organizations in Nigeria are becoming more aware of cybersecurity issues. Initiatives7 have been undertaken to educate the public and businesses about the inherent risks associated with cybercrimes and the necessity of implementing robust cybersecurity measures.

III. THE NEED FOR A RISK MANAGEMENT FRAMEWORK

Risk management is a systematic approach to identifying, assessing, and mitigating potential threats and vulnerabilities to protect an organization’s digital assets, data, and overall operations [2]. It involves a series of structured processes and measures aimed at safeguarding an organization’s information systems from a wide range of cyber threats, including hacking attempts, data breaches, malware infections, and other malicious activities [14]. [14] and [20] argue that RMF is indispensable for universities as it safeguards sensitive data, supports academic and research endeavors, fosters collaboration, and upholds trust in an era where cybersecurity threats are pervasive and evolving. Because universities often operate in open and collaborative environments, multiple users have varying degrees of access to the network [2]. This openness increases the attack surface, making it essential to systematically identify, assess, and manage cybersecurity risks [12].

A strong RMF maintains trust with students, faculty, and partners, protects data, enhances reputation, and demonstrates a commitment to cybersecurity best practices [23]. Therefore, the following were identified by [23], [8], [9], [6], and [22] for a robust RMF. The RMF typically includes steps such as risk identification, risk assessment, risk mitigation, risk monitoring, and continuous improvement. It also provides a strategic framework for organizations including universities to develop and maintain a proactive approach to cybersecurity, aligning it with business goals and regulatory requirements [14].



Figure 1- RMF components [8]

Identification of Risks. The first step in risk management is identifying and understanding the various risks that an organization’s digital environment faces [6]. This involves recognizing potential threats, vulnerabilities, and weaknesses that could be exploited by cybercriminals or malicious actors. Assessment of Risks. Proceeding risks identified, they are assessed in terms of their potential impact and likelihood [22]. Cybersecurity professionals evaluate the potential harm that could result from a successful cyberattack and the probability of such an attack occurring. Subsequently, [9] argued that this is one of the most crucial subsets of RMF and proposed a more narrowed aspect of risk assessment as a framework.

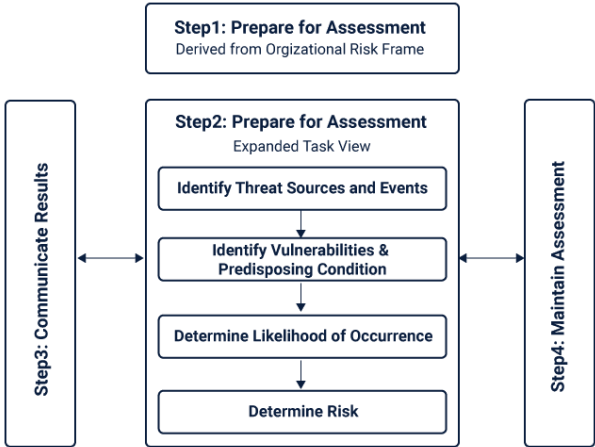


Figure 2-Risk Assessment Framework [RAF] [22]

[22] proposed that a RAF [Figure 2] a four critical approach to mitigate cybercrimes. This is shown in the table below.

TABLE 1- RAF AS PROPOSED BY [23] AND [22]

<i>Risk Assessment [Preparation] [RAP]</i>	<i>Conducting Risk Assessment [CRA]</i>	<i>Communicating Risk Assessment Results to Stakeholders [CRARS]</i>	<i>Maintaining Risk Assessments [MRA]</i>
Initial preparation and planning before conducting a risk assessment	Systematically identify potential risks, vulnerabilities, and threats that could impact the organization’s assets and operations.	Results sharing of identified risks and their severity with relevant stakeholders and decision-makers	Regularly reviewing and updating risk assessments
Defining the scope and objectives of the assessment	Assess the likelihood and potential impact of these risks.	Potential mitigation strategies communication	Regular updates to new threats, technology change, and organizational priorities shift
Identifying the assets and resources to be assessed and assembling the necessary resources, tools, and expertise		Clear and concise communication reports for informed decisions	Maintaining risk assessments ensures that the organization remains proactive in managing risks and adapting to new challenges

From Table 1, effective risk assessment is crucial for comprehensive risk management in cybersecurity [1]. It involves several key steps, starting with preparation, where objectives, scope, and necessary resources are defined [22]. The core phase is the assessment itself, where potential risks, vulnerabilities, and threats are systematically identified and evaluated using various methodologies and tools [23]. Clear communication of assessment findings, including identified risks and mitigation strategies, to relevant stakeholders and decision-makers is paramount for informed decision-making [22]. Lastly, risk assessment is an ongoing, evolving process, with regular reviews and updates to adapt to the changing threat landscape. These steps are fundamental in enhancing an organization's cybersecurity posture and overall risk management strategy [23].

Risk Prioritization. Not all risks are equal, and organizations must prioritize them based on their severity and potential consequences. This prioritization helps allocate resources effectively to address the most critical risks first [22].

Risk Mitigation and Controls. After identifying and assessing risks, risk management involves implementing measures, policies, and controls to reduce or mitigate those risks [22]. These controls may include firewall systems, intrusion detection systems, encryption, employee training, and incident response plans.

Continuous Monitoring. Effective risk management is an ongoing process. Organizations continuously monitor their systems and networks for potential threats and vulnerabilities [23]. They also review and update their risk management strategies to adapt to evolving cybersecurity landscapes.

Incident Response. Despite preventive measures, incidents may still occur. Risk management includes having robust incident response plans in place to quickly detect, respond to, and recover from cyber incidents. This helps minimize damage and downtime [23].

Compliance and Reporting. Organizations often need to comply with cybersecurity regulations and standards [23]. Risk management includes ensuring that the organization's cybersecurity practices align with these requirements [23, 12]. Additionally, reporting mechanisms are established to communicate the state of cybersecurity risk to stakeholders [13].

[22] and [2] agree that cybersecurity RMF is a holistic approach that aims to proactively identify, assess, and address potential threats and vulnerabilities in an organization's digital environment. Therefore, as a dynamic and ongoing process, it is essential in protecting an organization's assets and operations in an increasingly digital and interconnected world.

IV. PROPOSED RISK MANAGEMENT FRAMEWORK (RMF) FOR NIGERIAN UNIVERSITIES

A. The Role of Artificial Intelligence in Cybersecurity

Artificial Intelligence [AI] plays a pivotal role in cybersecurity by enhancing threat detection and response [24]. According to [25], AI-driven systems can analyze vast datasets in real-time, identifying anomalies and potential security breaches more effectively than traditional methods. [24] adds that machine learning models can adapt and evolve to detect new and evolve threats, providing proactive defense.

AI-powered security solutions can also automate routine tasks, freeing up human experts for more complex analysis and decision-making [25, 16]. However, [26] argues that the same AI technology can be weaponized by cybercriminals to launch sophisticated attacks, highlighting the need for a continuous arms race between defenders and attackers in the ever-evolving cybersecurity landscape [23].

As proposed by [25], AI's ability to enhance threat detection, automate risk assessment, and predict emerging risks makes it a valuable component of a modern RMF for cybersecurity. Certain capabilities enable organizations to proactively manage and mitigate cybersecurity risks effectively [24]. Firstly, [27] argues that AI contributes to threat detection and mitigation within RMFs. Machine learning algorithms can analyze large volumes of data, including network traffic, to identify anomalies and potential security threats. This proactive threat detection helps organizations respond swiftly to emerging risks [27].

Secondly, [24] adds that AI-powered automation can streamline risk assessment processes within RMFs. Universities' AI-driven risk assessment tools for instance, can continuously monitor its digital environment, identify vulnerabilities, and prioritize risks based on potential impact and likelihood. This assists in allocating resources efficiently for risk mitigation [24]. Furthermore, [27] described AI-driven predictive analytics ability to forecast potential cybersecurity risks, allowing organizations like universities take preemptive measures. Therefore, by analyzing historical data and identifying trends, AI can help universities anticipate and prepare for emerging threats, reducing the likelihood of security incidents [25, 26, 15, 24].

A typical example as described by [27] is the User and Entity Behavior Analytics [UEBA]. UEBA solutions leverage AI and machine learning to monitor user and entity activities across an organization's network. In educational institutions, UEBA can detect anomalies in user behavior that may indicate a security threat [27]. In a situation where a student's account suddenly starts accessing sensitive data or resources at unusual times or from unfamiliar locations, the UEBA system can flag this as suspicious activity [Ibid]. It can also identify compromised accounts or insider threats by analyzing patterns of behavior [28]. Therefore, by constantly analyzing data from various sources, UEBA systems provide early warnings of potential cyber threats, allowing educational institutions to take prompt action to mitigate risks [28]. This proactive approach is crucial for protecting sensitive student and faculty data and maintaining the integrity of educational processes. Hence, [27] reiterated the AI-powered UEBA systems as a valuable tool in cybersecurity for education institutions with the ability to enhance threat detection and response capabilities, helping institutions proactively address cybercrime and safeguard their digital environments.

B. The Role of RMF In Mitigating Cyberthreats and Cybercrimes

As aforementioned, RMFs play a significant role in mitigating cyberthreats and cybercrimes by providing a structured and comprehensive approach to identifying, assessing, and managing cybersecurity risks [1, 12, 20]. These frameworks are essential for organizations, including

educational institutions, to protect their digital assets and sensitive data from evolving cyber threats [1]. [12] argues that RMFs facilitate the systematic identification and assessment of cybersecurity risks [a crucial first step in RMF structure8]. Thus, by categorizing and prioritizing potential threats, universities can better understand their vulnerabilities and the potential impact of cyberattacks. This information forms the foundation for implementing targeted security measures. For example, the National Institute of Standards and Technology [NIST] Cybersecurity Framework provides a structured approach to assess risks and prioritize protective actions⁹.

Post-identified risks, [2] proposed that RMFs can be used to deploy security controls and safeguards to protect against specific threats. For instance, the Center for Internet Security [CIS] provides a set of critical security controls that organizations can implement to address common cyber threats effectively [29]. Thus, with continuous monitoring, RMFs can assist an organization's cybersecurity posture [25]. This proactive approach involves regularly assessing the effectiveness of security measures, identifying weaknesses, and responding promptly to emerging threats, ensuring ongoing cybersecurity resilience [29].

Since RMF incorporates regulatory and compliance requirements, universities and other organizations can meet legal and industry-specific cybersecurity obligations [28]. Albeit this not only safeguards against cybercrimes but also establishes accountability in case of security breaches especially when robust incident response and recovery plans are in place. According to [29], in the event of a cybercrime or cybersecurity breach, predefined protocols are followed to minimize damage, contain the incident, and swiftly recover operations. Therefore, frameworks like the Incident Command System [ICS]¹⁰ are instrumental in managing emergency situations, including cybersecurity incidents [30]. According to [18], user awareness and training are paramount within RMFs. Therefore, [18] proposed that since RMFs provide a holistic approach to cybersecurity risk management [protecting digital assets and mitigating cyberthreats], minimizing the impact of cybercrimes is possible when effectively implemented not only to enhance security but also demonstrate a commitment to cybersecurity best practices and compliance with relevant standards and regulations.

C. Useable RMF For Nigerian Universities

There have been several proposed RMF structures useful in mitigating cybercrimes in educational institutions and other institutions, however, Figure 3 is an example of an RMF incorporated cybersecurity mitigation approach for Wellspring University, Edo state, Nigeria. Although the approach encompasses a pre-cyberattack, cyberattack management and post-cyberattack management, it may be more comprehensible to specifically redefine this RMF structure and incorporate the standard recommended structure by the National Institute of Standards and Technology [NIST] Cybersecurity Framework. This is depicted in Figure 1.

However, depending on the budgeted funding available for RMF approaches, universities may be unable to effectively

mitigate cyberthreats and cybercrimes when there are defined, semi-defined and undefined elements in their RMF approach.

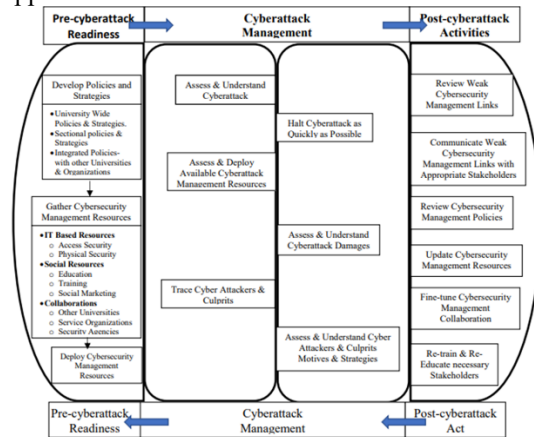


Figure 3- Wellspring University's RMF Structure and Comprehensive Approach to Cybersecurity [31]

For instance, the NIST Cybersecurity Framework as shown in Figure 4 is depicted in Figure 1, clearly defining all four steps. As proposed by [22] in Figure 2, the RAF is an 'undeniable' critical aspect of a well-implemented RMF structure. Therefore, based on the findings of this study, the NIST Cybersecurity Framework is recommended to Nigerian universities to ensure that they are prepared for future cyberattacks and threats. However, a well-implemented RMF may be hampered by resource constraints, complex IT environments, and user behavior impact cybersecurity efforts [32]. Legacy systems and privacy concerns add complexity [15], while evolving threats demand continual adaptation [17].

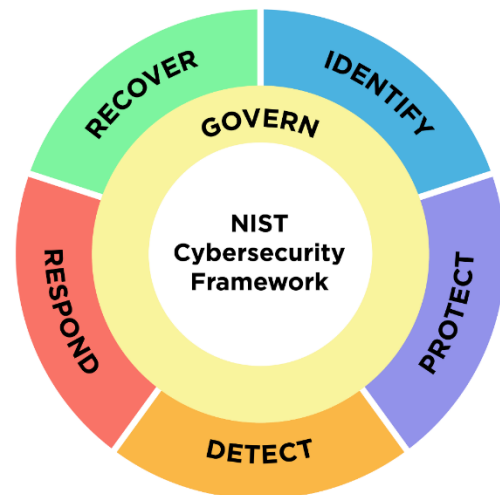


Figure 4- Cybersecurity Framework [CSF] 2.0 [Draft Version] by NIST [32]

V. DISCUSSION AND CONCLUSION

Two limitations were identified in the study. The study's niche was centered on Nigerian universities and may also be non-generalizable because of the country's uniqueness in culture, differences in university structure, and government

regulations. Secondly, RMF is multifaceted— involving technical, legal, organizational, and educational aspects, making it a complex subject to research comprehensively. This was also identified by [5, 23, 14, 13, 7].

Numerous benefits come with embracing RMF in universities' systems to bolster cybersecurity. Nigerian universities can witness a substantial improvement in their security posture when they deploy sophisticated cybersecurity tools such as intrusion detection systems, firewalls, and encryption mechanisms. However, this heightened technical sophistication also demands a well-trained IT [information technology] workforce capable of continuously monitoring network traffic, promptly applying patches, and managing advanced security technologies. Furthermore, a critical technical implication lies in safeguarding data, particularly sensitive academic and student information. This involves the meticulous implementation of encryption and access control measures, alongside rigorous data backup protocols.

On one hand, Nigerian universities may need to establish robust technical incident response systems, potentially investing in specialized tools and forming dedicated incident response teams. Adequate budget allocation for cybersecurity initiatives is non-negotiable. On the other hand, universities must earmark resources for staff training, software licenses, hardware upgrades, and ongoing maintenance. Cybersecurity awareness and training/ for non-technical staff and stakeholders are equally vital. Creating a culture of cybersecurity necessitates comprehensive training, effective communication, and the cultivation of a shared understanding of cybersecurity principles. Moreover, regulatory compliance with evolving data protection laws constitutes a critical non-technical aspect. Aligning university policies with Nigerian data protection regulations may require legal consultations to ensure full compliance. This also opens room for more implementation of cybersecurity measures that can disrupt traditional workflows, making effective change management imperative to help non-technical staff adapt to new security protocols without hampering productivity.

None of these can be executed without effective stakeholder communication. While conveying cybersecurity incidents and outcomes to stakeholders is non-technical, it's essential for building trust and transparency. However, this task entails translating technical details into easily understandable terms to ensure all stakeholders can comprehend the implications and responses to cybersecurity incidents.

Recommendations

By implementing these recommendations, Nigerian universities can significantly enhance their cybersecurity posture and develop a robust Risk Management Framework to protect sensitive data, uphold academic integrity, and ensure the continuity of critical operations.

Comprehensive Cybersecurity Policy and Governance

It's crucial to have a solid cybersecurity policy in place and to set up a specialized cybersecurity governing body to oversee it. This policy should outline objectives, roles, responsibilities, and compliance with regulations.

Risk Assessment, Training, and Collaboration

Regularly assessing risks is crucial to identifying vulnerabilities that could compromise the security of your organization. Additionally, investing in training and awareness programs for your staff and stakeholders can go a

long way in ensuring that everyone is equipped with the knowledge and skills to keep your organization safe. Collaborating with cybersecurity organizations to stay updated on emerging threats and best practices is also necessary.

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